Echolabel: Image-based echogram annotation tool for aquatic ecology

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Abstract

Active acoustic systems are essential for studying aquatic ecosystems, yet processing these data still relies heavily on manual annotation, and to our knowledge no open-source software exists for this task. We present Echolabel, an open-source, cross-platform tool for fast echogram visualization and annotation, producing CSV outputs compatible with Echoregions, an annotation package in Echostack. Deliberately minimal and task-specific, it offers a command-line interface and Python API and builds on the LabelMe interface to annotate cached echogram images, converting between image and physical (time, depth) coordinates. Visualization includes RGB-composite echograms. We demonstrate the tool's utility with Echoregions on routine annotation tasks and a brief ecological analysis, and show that it is suitable for delineating irregular, fuzzy objects such as sound-scattering layers.

Keywords: Active acoustics, Multifrequency, Echogram annotation, Aquatic ecology, Open source

Metadata

Nr	Code metadata description	Metadata
C1	Current code version	V0.1.0
C2	Permanent link to code/repository used for this code version	https://github.com/gaspardringuenet/echolabel
C3	Permanent link to reproducible capsule	https://doi.org/10.5281/zenodo.24088063
C4	Legal code license	GPL-3.0 license
C5	Code versioning system used	Git
C6	Software code languages, tools and services used	Python
C7	Compilation requirements, operating environments and dependencies	Requires Python $\geq$ 3.12

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Nr	Code metadata description	Metadata
C8	If available, link to developer documentation/manual	https://github.com/gaspardringuenet/echolabel/README.md
C9	Support email for questions	gaspard.ringuenet@umontpellier.fr

1. Motivation and Significance

Aquatic ecosystems, from lakes and rivers to coastal waters and the open ocean, are largely hidden from direct observation. Many organisms live in the water column, which nets and traps sample sparsely, at high cost and often invasively. Active acoustic systems (sonars) provide crucial information: by sounding the water column at high resolution, they are the only technology capable of continuously observing organism distribution and behaviour at the spatial and temporal scales of physical and ecological processes [1].

Although this tool applies to echograms from any aquatic environment, the examples come from the pelagic ocean. The pelagic zone is the water column away from the seabed. This vast habitat represents over 90% of the planet's living volume. Unlike the benthic zone (i.e., the seabed) or land, it lacks hard physical structures and can be divided into depth layers based on light penetration. The mesopelagic, where less than 1% of light reaches, typically spans 200–1000 m. Its significance is evident from its large biomass and diversity [2, 3, 4], crucial role in ocean functioning [5, 6], and interdependence with human societies [7, 8]. Despite this, the "twilight zone" remains understudied, primarily due to technical limitations in investigating deep waters. Active acoustics is particularly valuable here, as echosounders can sample the water column to depths of ~1000 m at high resolution.

However, extracting ecological information from acoustic data remains a major challenge. Manual delineation of features in echograms has long been a common practice in marine acoustics. Targeted biological features are most often aggregations such as fish schools or krill swarms, but may also include sound-scattering layers comprising organisms from invertebrates to fish distributed horizontally in the water column.

Annotating these features is part of the daily routine in ecosystem acoustic surveys, whether for subsequent analyses of distribution and biomass or to populate libraries used to train machine learning models for species identification. Several software solutions are currently available for this purpose: commercial options such as Echoview [9]; free but closed-source Windows-based software such as LSSS [10] and Movies3D [11]; or open-source software requiring a proprietary MATLAB license, such as ESP3 and Matecho [12, 13].

Free, open-source, cross-platform ecosystem-acoustics software has developed significantly in recent years. The most notable example, in both development and scientific-community engagement, is Echostack, a suite of Python packages for routine tasks with scientific echosounder data [14, 15]. Echostack includes the Echoregions package, designed to parse acoustic annotations and interface them with gridded acoustic data. The package contains functions to create masks for machine-learning model training and routine processing tasks such as biomass estimation. However, no open-source solution is yet available for such annotations, although annotation is one of the tasks acoustic operators spend the most time on and the first step in ecosystem acoustic processing.

To bridge this gap, we propose Echolabel, a minimalist Python application for echogram annotation, interoperable with Echostack and Echoregions. Unlike most GUI-based acoustic software, which provides suites of analysis-ready tools, Echolabel focuses solely on annotation for open-source acoustic-processing pipelines. It leverages LabelMe [16], a widely used open-source Python image-annotation application, by converting acoustic data into echogram images. The output is an Echoregions-compatible CSV file for integration into Python acoustic-processing workflows. Echolabel is free, open-source, cross-platform, and has no proprietary dependencies. It requires little computing power and offers an easy-to-use GUI for annotation.

2. Software description

2.1. Software architecture

The software is written in Python, with separate modules responsible for application configuration, command-line inputs, cached data management, and parsing annotation files.

It comprises 14 Python modules of roughly 100–200 lines each and avoids reimplementing existing functionality. The annotation interface is provided entirely by LabelMe v6.0.0 [17], while Echolabel parses annotations between LabelMe and Echoregions. The parsing module, `regions2d_extension`, mimics Echoregions' code for parsing Echoview annotation data, clarifying the application's purpose.

Echolabel has been developed to be primarily used from the command-line interface (CLI; see Listing 1). The main `process` command triggers conversion of the acoustic data into images before automatically running LabelMe. It features a number of optional parameters, allowing the user to modify the echogram. Passed CLI parameters are used to override a default configuration file, which can be directly edited for improved control and reproducibility. If necessary, a programming interface remains available with the `EcholabelApp` and `Config` classes at the top level.

A series of unit tests ensure that core features are preserved across future updates. Tested functionalities include configuration, data loading, image-dataset management, and annotation parsing. Git is used for version control.

2.2. Software functionalities

Echolabel wraps LabelMe, handling shape-coordinate conversion. It loads acoustic data, chunks it into echogram images, and runs LabelMe. When the user exits the GUI, image-based annotations are converted to physical (time, depth) coordinates and saved in a format readable by Echoregions.

Echolabel works with processed, gridded volume backscattering strength data (Sv, dB) [18], i.e., the widely used acoustic measure for characterizing the distribution of organisms in the water column, after the data have been cleaned of artifacts and regridded onto a consistent grid across frequency, depth, and time–distance dimensions. By default, Echolabel supports two widely used conventions for distributing data with this structure and level of processing: the IMOS SOOP-BA NetCDF data format [19] and the coordinate/variable names adopted by Echostack (namely using "ping_time" instead of "time" as time dimension coordinate name). For full flexibility, it can support any variable names, provided that the user manually identifies Sv, depth, and time–distance variables in the configuration file. Both NetCDF files and Zarr stores are accepted as input formats. Several files in the same directory may be loaded as a single dataset as long as their coordinates are compatible.

Users can select the visualized frequency, value bounds, and colormap by name. In addition to single-frequency echograms, Echolabel offers a triple-frequency RGB mode, mapping three frequencies to the red, green, and blue channels with equal bounds. This visualization helps assess frequency-response distributions across the echogram (see examples in [20, 21]). The background colour for inactive pixels is configurable as a matplotlib-compatible string or tuple.

Running the application triggers data loading and the building and caching of an echogram image dataset using visualization options. Echogram images are the representation of acoustic data once colour mapping has been applied. Each pixel represents a sampled volume. Each image's physical (time and depth) coordinates are saved in cache. Image-dataset construction is the only computationally intensive step. Although lazy-loading prevents memory overload, construction may take a few minutes for large surveys with high-resolution data. Cached images can be reused between sessions, allowing the GUI to launch almost instantaneously.

A prominent feature of Echolabel is the LabelMe graphical user interface (GUI; Figure 1), which is used to create annotations. This relies on the prior conversion of the acoustic data into images. Within the open window, the user can loop through the images in the "File List" panel. They can also visualize and edit existing annotations using the "Annotation List" panel and point clicks. Most importantly, they can use the

"CreateShapeTools" to add new shapes. All changes are automatically saved in cache as LabelMe JSON annotations during the GUI editing session. The GUI is customizable, and the user has the option to edit the LabelMe configuration file on their system. Since Echolabel runs LabelMe as a subprocess and only operates on annotation files, modifying the LabelMe configuration should not create conflicts.

When closing the GUI, shape annotations are parsed and saved as a CSV file compatible with the `echoregions.Regions2D` class. This conversion uses the saved image coordinates to convert pixel-wise annotations into (time, depth) points. `Regions2D` objects can represent polygons only. For this reason, polygon and rectangle LabelMe tools are the only ones whose shapes can be converted (output from other tools is ignored). Once the annotation-format conversion is done, the CSV file is created or updated, and is then ready to be loaded into Echoregions in a single line.

Additional features include a `demo` command that downloads and processes demonstration data. It is also worth noting that LabelMe provides an AI polygon creation tool. The tool installs an image segmentation model (Meta's Segment Anything Model 2 as of LabelMe v6.0.0) locally to provide assistance by automatically delineating objects. While such models are efficient on natural images, they appear much less useful on echograms. However, since the tool produces polygons, annotations are correctly parsed by Echolabel.

3. Illustrative examples

3.1. Polygon annotation on RGB echogram

The ABRAÇOS II survey was conducted off the north-eastern coast of Brazil during the Austral fall of 2017 (April 9th–May 7th). Research vessel Antea of the French oceanographic fleet operated a hull-mounted EK60 echosounder at four frequencies (38, 70, 120, and 200 kHz) with a pulse length of 512 s [22, 23]. Two configurations were used: a ping rate of 0.45 s for seabed depths below 150 m and 2.4 s for deeper areas, recording to depths of 750 m. The echosounder was calibrated mid-survey in Fernando de Noronha using standard protocols [24]. Acoustic data were processed using Matecho. Noise and artifacts were removed using automated and manual cleaning tools. Volume backscattering strength was echo-integrated with a minimum threshold of -90 dB in 3-ping by 1-m cells. The echo-integrated product was converted from Matecho's MATLAB format to a Zarr store. The demonstration data are a 10-hour subset of the ABRAÇOS II survey, spanning open ocean and part of the Fernando de Noronha island shelf, up to ~50 m deep.

Several objects of interest are selected from the demonstration dataset using Echolabel. A simple terminal command launches the GUI and renders an RGB composite echogram showing the 38, 70, and 120 kHz channels simultaneously (Listing 1). Selected features are delineated using the "Create Polygons" and "Create Rectangles" tools and assigned text labels (Figure 1): a sound-scattering layer ("Layer"), a probable fish aggregation ("School"), a v-shaped anomaly probably caused by a lowered instrument ("Artifact"), and rectangular areas of varying hue ("Unknown"). The eight annotated regions are saved as a CSV file when the GUI closes.

Listing 1 Example bash command to build an RGB echogram and run the ROI demonstration.

```
echolabel process \
-s data/demo.zarr \                # acoustic data store
-o data/demo_regions.csv \         # annotation file
-f 38000 -f 70000 -f 120000 \      # mapped frequencies (Hz)
--cmap RGB \                       # RGB composite echogram
--vmin -90 --vmax -50              # Colour scale bounds
```

Echolabel solely focuses on creating the annotations, but the software's strength lies in its immediate compatibility with the Echoregions package. The following sections therefore demonstrate first-order use cases for such annotations using Echoregions.

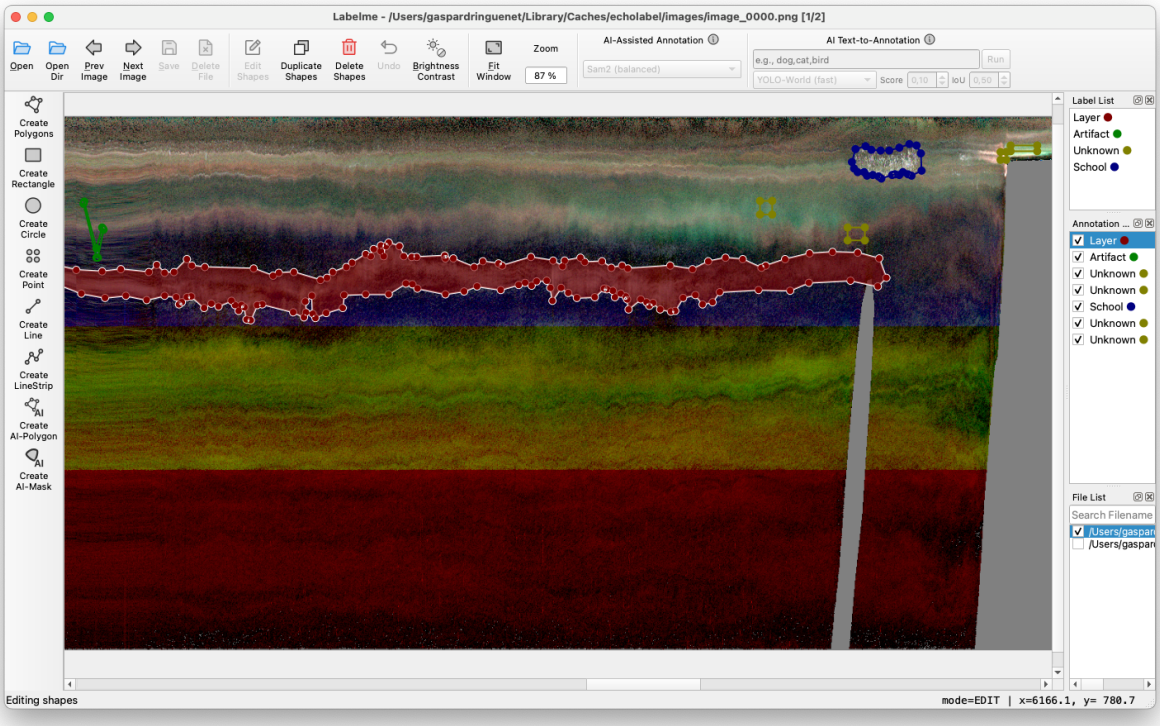


Figure 1: LabelMe graphical user interface showing drawn regions on an RGB composite echogram (R: 38 kHz; G: 70 kHz; B: 120 kHz, as specified in Listing 1) in the image panel (centre). The elongated grey shape masks double-echo data. The second grey shape on the right indicates pixels below the ocean floor. Horizontal colour boundaries result from the channels' range limits (e.g., the 120 kHz channel becomes too noisy around 250 m, so data are removed below that depth). Manually created polygonal shapes are overlaid on the echogram. Their labels are visible in the "Annotation List" panel and as a non-duplicated list in the "Label List" panel. A "Layer" annotation and its polygon are being edited and are therefore highlighted. The "File List" panel shows all available echogram images.

3.2. Use case 1 - Routine tasks

Figure 2 shows regions with the "Unknown", "Artifact", and "School" labels (a), along with processing or data analysis steps corresponding to the underlying echogram features (b-g): frequency response investigation, artifact removal, and aggregation mask extraction, respectively. These tasks are enabled by Echoregions' masking functionality (using the regionmask package), and each requires no more than a dozen lines of code (excluding figure construction).

The most basic task is computing the frequency responses of an echogram region. The influence of the organisms' anatomy on their back-scattering properties has long been acknowledged, and is the motivation for multifrequency acoustics [1]. Spatio-temporal variations in frequency responses can then be attributed to variations in the species (or age) composition in the sampled volumes. While the RGB echogram successfully renders part of this contrast, a quantitative comparison of back-scattering between regions of interest provides a more exhaustive and objective insight. Figure 2 b,c compare two reddish (38 kHz-dominant) and two greenish (70 kHz-dominant) regions across oceanographic settings (pelagic or shelf edge), depths (150 m or 50 m), and intensities (apparently stronger near the shelf). We compute frequency response relative to the 38 kHz channel ($R(f) = \Delta\text{MVBS}_{f-38\text{kHz}} = S_v(f) - S_v(38\text{ kHz})$; see [18] for notation). Green regions have similar R(f) values, whereas red regions differ substantially, especially at 120 and 200 kHz, suggesting different biological compositions. Further analysis using this kind of regions of interest may include target strength (TS) modelling, or building a training set for a multifrequency classification algorithm.

Despite a thorough cleaning of acoustic data, some unwanted signal or noise may still be present in the final product. This is accentuated by the context-dependence of unwanted signal. Manual cleaning of the echo-integrated product can easily be conducted with the combination of Echolabel and Echoregions. This is demonstrated in Figure 2 d,e in which the back-scattering of a probable CTD profiler is removed using a polygonal shape.

On the contrary, one may want to extract the mask of a biological aggregation of interest. This is common for fish schools and shoals, and illustrated in Figure 2 f,g. A typical use of such masks is supervised deep learning model training. Computation of acoustic variables, or post processing of the mask (for instance to remove unwanted pixels) may also apply to fish aggregations.

3.3. Use case 2 - Analysis of a sound-scattering layer

A wide variety of data analyses can be conducted using masked echograms. In this section we propose to illustrate how Echolabel, together with Echoregions, can be used to test a simple hypothesis. A close inspection of the data within the "Layer" polygon on the RGB echogram shows an interesting pattern. The layer varies both in depth, with finer-scale variations superimposed on the diel vertical migration pattern, and in frequency response, with some sections appearing brown while others appear grey (most visible in Figure 2 a). We hypothesize that changes in frequency response are linked to changes in the mean tilt angle of mesopelagic organisms (presumably fish in this case). To test this, we compare $\Delta\text{MVBS}_{70-38\text{kHz}}$, to the vertical speed of the layer (positive downward), assuming vertical speed is tightly linked to the average tilt angle of the fish. Since the vessel was moving during data acquisition, we must also make the assumption that the observed variations in the layer depth are caused by organisms moving vertically in the water column, and not due to spatial heterogeneity.

The layer is delineated manually using Echolabel, as shown in Figure 1, and the corresponding mask is generated with Echoregions. The irregular and fuzzy nature of the extracted object calls for manual extraction, making this an exemplary ecological use case for our software. For each timestamp, the mean depth of the layer is computed, weighted by the nautical area scattering coefficient (NASC [$\text{m}^2\ \text{nmi}^{-2}$], a linear measure of the back-scattered energy [18]). The computed line is smoothed using a rolling mean spanning 50 timestamps. From this vector, the layer's downward migration speed is obtained by differentiation with respect to time. The depth-averaged $\Delta\text{MVBS}_{70-38\text{kHz}}$ is visually compared to both migration depth and speed (Figure 3). A linear regression is used to test the dependence of frequency response on vertical speed.

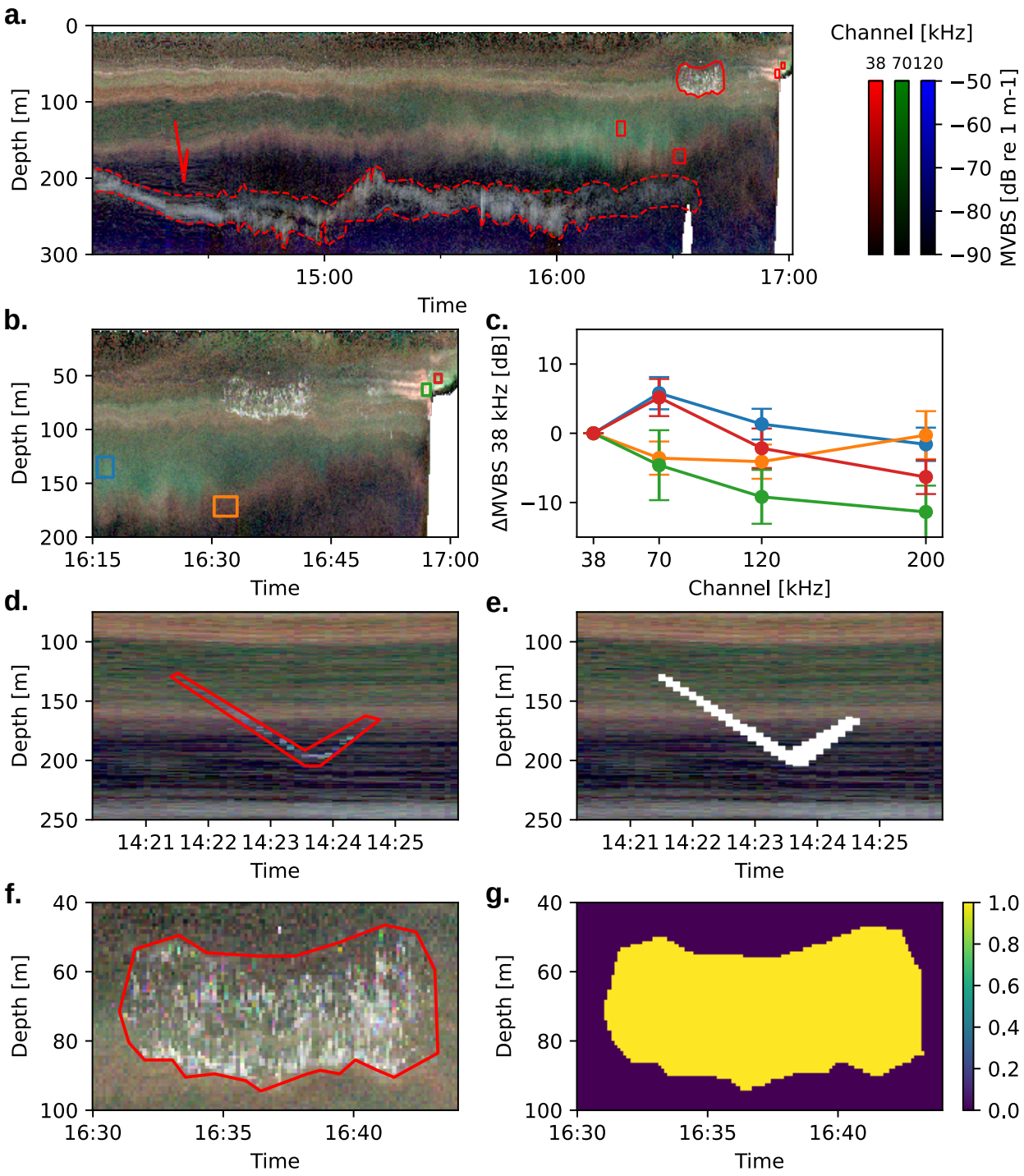


Figure 2: Annotated echogram regions and simple use cases. (a) Annotated polygonal regions displayed on top of an RGB echogram, with the dashed line representing the "Layer" polygon. (b-g) Use cases: the annotated rectangles (b) can be used to compute the mean and standard deviation of relative frequency responses of data within (c); a v-shaped anomaly, typical of a lowered instrument (d) can be masked out (e); the binary mask (g) of a fish aggregation (f) can be computed.

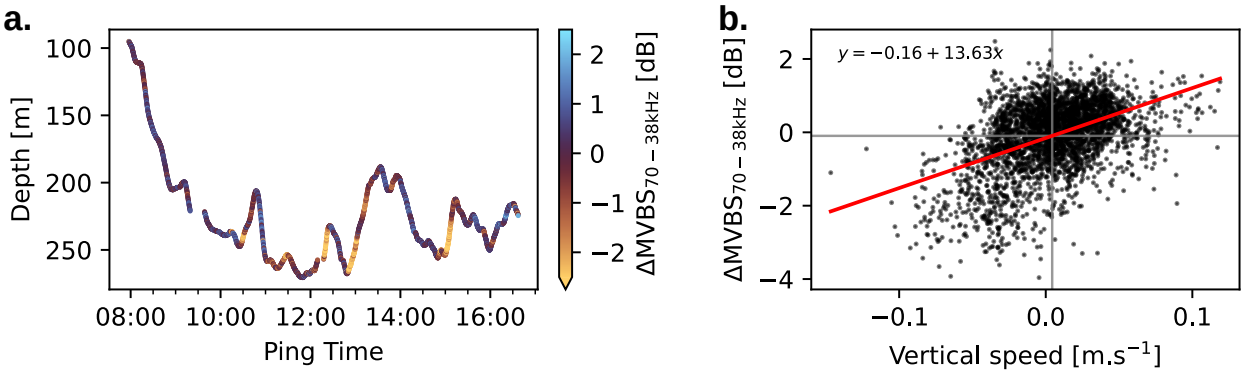


Figure 3: Example first-order analysis of a diffuse sound-scattering layer (selected as the "Layer" polygon in Figure 1). (a) Time series of the layer's center-of-gravity depth (NASC-weighted mean depth), smoothed with a 50-timestamp rolling mean. Colour indicates the sign and magnitude of the relative frequency response. (b) Scatter plot of $\Delta\text{MVBS}_{70-38\text{kHz}}$ against downward vertical speed. The red line shows the linear regression fit; the fitted equation appears in the top left corner. Grey lines indicate the means of both variables.

Figure 3 a supports the visual intuition given by the RGB echogram. Marked upward-swimming events correspond to a higher response at 38 kHz than at 70 kHz. These are the only occurrences of extreme (-4 to -2 dB) $\Delta\text{MVBS}_{70-38\text{kHz}}$, while most points have smaller frequency-response differences. This is consistent with the scatter plot in Figure 3 b, where most data fall within a small range of $\Delta\text{MVBS}_{70-38\text{kHz}}$ (-1 to 1 dB) and vertical speed (-5 to 5 cm s^{-1}). A minority of points show extreme values of both variables, with concordant signs (both negative or both positive). Even accounting for the near-zero points, the linear model shows a positive dependence of $\Delta\text{MVBS}_{70-38\text{kHz}}$ on downward speed ($p < 10^{-3}$), with satisfactory residual distributions.

This exploratory analysis identifies a correlation between vertical speed and $\Delta\text{MVBS}_{70-38\text{kHz}}$ and shows their statistical dependence. Under our assumptions, this result is consistent with the tilt-angle hypothesis: frequency-response variations are associated with fish movements rather than changes in species composition, although the latter cannot be ruled out. The correlation also suggests that we observe organism speed rather than spatial variability, supporting both assumptions. The stronger effect during swift upward-swimming events (visible in Figure 3 a at 12:30, 13:00, and 15:00, and in the poorer model fit for small values in Figure 3 b) could indicate that these events are more coordinated than downward migrations, with more animals aligned and swimming in the same direction. We also observe that the organisms eventually return to a depth consistent with a first-order u-shaped diel vertical migration.

Light level is the main driver of diel vertical migration, with organisms seeking refuge in deeper waters during the day to avoid visual predators [25]. Yet, during some observed upward-swimming events, the layer raised more than 50 m above its nominal depth (Figure 3 a). The sudden and coordinated nature of these events could suggest that the layer is reacting to a threat [26] or avoiding a physical barrier. Another explanation could be weather events changing skylight intensity, as animals may also avoid darkening waters. Further analysis and evidence are required to address these questions properly. This analysis nonetheless illustrates the potential of manual acoustic data annotation, and so of Echolabel, to address questions in pelagic behavioural ecology.

4. Impact

Echolabel bridges a gap for aquatic scientists using Python who want to integrate acoustic annotation into Echoregions-dependent pipelines. The flexible acoustic dataset loading framework allows most acoustic NetCDF conventions to be loaded, and the use of Echoregions' CSV convention for annotated shapes enables subsequent analysis with the package. In addition, relying on cached echogram images makes the annotation

process responsive on most machines. Together with the smooth and fast LabelMe GUI, this enables efficient echogram annotation. The RGB composite mode also makes frequency-response-aware labelling possible.

Echolabel is a single-purpose tool, and is not intended to replace acoustic processing software. This tool was primarily developed for an internal purpose. Its core functionality, i.e., annotating polygonal shapes on echograms, can be achieved using most general-purpose GUI-based acoustic software (although the use of RGB echograms remains relatively uncommon among them). However, the outputs of few such software packages can currently be parsed by Echoregions.

Nevertheless, we believe that a minimalist and fast tool might be of use to the community. It is particularly well-suited to individual researchers wanting to annotate a large number of echogram features that do not require inspecting data/metadata values during annotation (image files do not provide such information; such information can, however, be accessed during subsequent processing with Echoregions).

Future development of open-source acoustic tools might render Echolabel obsolete. Echoregions might extend its supported annotation formats. Other tools might arise for the annotation of dynamic echograms, possibly in Python using datashader-based mechanisms to reduce the computational cost of rendering. A shallow layer over the HoloViews library might be a suitable approach, allowing users to create custom interfaces easily using Panel. Such developments are needed to support a comprehensive Python stack for acoustic data analysis. To our knowledge, Echolabel remains the only open-source echogram annotation software directly compatible with Echoregions.

5. Conclusions

Echolabel provides a simple command-line interface for echogram annotation, producing outputs that integrate into Echostack, specifically the Echoregions package, a Python ecosystem for processing and analysing active acoustic data that is expected to become a standard in the field.

Our tool leverages a widely used image annotation software package by converting acoustic data into echogram image files and using LabelMe to annotate them. NetCDF and Zarr formats are accepted, and most conventions can be used as input (IMOS SOOP-BA and Echopype conventions are accepted by default). No innovation is claimed with respect to echogram generation: the software provides either basic single-channel mapping or tri-frequency RGB echograms.

Example use cases illustrate the ease with which various echogram features can be selected, and how the underlying data may be processed or analysed with minimal effort using the masking functionality of Echoregions after annotation with Echolabel. Such manual annotation proves particularly useful for fuzzy objects with unclear boundaries, which are commonplace in active acoustics (e.g. sound-scattering layers). Aside from convenience, Echolabel also enables very fast annotation workflows. However, the manual nature of the annotation task does not allow an objective measurement of performance, which is why no performance example was proposed in this work. Nevertheless, the absence of rendering cost after generating images, together with the widespread use of LabelMe within the computer vision community, supports the practical usability of the approach.

The scope of Echolabel is deliberately limited to echogram-image generation and annotation. Its image-based design entails limitations, including loss of direct access to data and metadata values and division of the dataset into individual frames. LabelMe also renders images using bilinear interpolation, causing pixel-scale blurring that may affect the precision of fine-scale annotations.

While Echolabel has clear limitations, it provides an efficient and convenient solution for tasks requiring the rapid annotation of large numbers of shapes. Future developments in the open-source active acoustics community may eventually provide more comprehensive alternatives, but Echolabel currently offers a lightweight, cross-platform, and Echoregions-compatible solution for this specific use case.

Availability

Echolabel is open-source and available at https://github.com/gaspardringuenet/echolabel. Additional source code, scripts and data used to demonstrate Echolabel are available at this manuscript's quarto project repository (https://github.com/gaspardringuenet/echolabel-manuscript).

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Artificial intelligence (AI) use statement

AI-based tools were used throughout software development to assist with planning, code review, improvement suggestions, debugging and writing tests. All AI-written source code has been reviewed. AI tools were used for review and trimming during manuscript preparation, and for clarity improvement in the abstract section.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] K. J. Benoit-Bird, G. L. Lawson, Ecological insights from pelagic habitats acquired using active acoustic techniques, Annual Review of Marine Science 8 (1) (2016) 463–490. doi:10.1146/annurev-marine-122414-034001.
URL https://www.annualreviews.org/doi/10.1146/annurev-marine-122414-034001

[2] X. Irigoien, T. A. Klevjer, A. Røstad, U. Martinez, G. Boyra, J. L. Acuña, A. Bode, F. Echevarria, J. I. Gonzalez-Gordillo, S. Hernandez-Leon, S. Agusti, D. L. Aksnes, C. M. Duarte, S. Kaartvedt, Large mesopelagic fishes biomass and trophic efficiency in the open ocean, Nature Communications 5 (1) (2014) 3271. doi:10.1038/ncomms4271.
URL https://www.nature.com/articles/ncomms4271

[3] L. Eduardo, M. M. Mincarone, T. Sutton, A. Bertrand, Deep-pelagic fishes are anything but similar: A global synthesis, Ecology Letters 27 (9) (2024) e14510, _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ele.14510. doi:10.1111/ele.14510.
URL https://onlinelibrary.wiley.com/doi/abs/10.1111/ele.14510

[4] S. H. D. Haddock, C. A. Choy, Life in the midwater: The ecology of deep pelagic animals, Annual Review of Marine Science 16 (Volume 16, 2024) (2024) 383–416. doi:10.1146/annurev-marine-031623-095435.
URL https://www.annualreviews.org/content/journals/10.1146/annurev-marine-031623-095435

[5] C. Robinson, D. K. Steinberg, T. R. Anderson, J. Arístegui, C. A. Carlson, J. R. Frost, J. Ghiglione, S. Hernández-León, G. A. Jackson, R. Koppelmann, B. Quéguiner, O. Ragueneau, F. Rassoulzadegan, B. H. Robison, C. Tamburini, T. Tanaka, K. F. Wishner, J. Zhang, Mesopelagic zone ecology and biogeochemistry – a synthesis, Deep Sea Research Part II: Topical Studies in Oceanography 57 (16) (2010) 1504–1518. doi:10.1016/j.dsr2.2010.02.018.
URL https://linkinghub.elsevier.com/retrieve/pii/S0967064510000846

[6] U. Passow, C. A. Carlson, The biological pump in a high $co_{2\ world}$, Marine Ecology Progress Series (2012). doi:10.3354/meps09985.
URL https://sci-hub.fr/10.3354/meps09985

[7] A. Martin, P. Boyd, K. Buesseler, I. Cetinic, H. Claustre, S. Giering, S. Henson, X. Irigoien, I. Kriest, L. Memery, C. Robinson, G. Saba, R. Sanders, D. Siegel, M. Villa-Alfageme, L. Guidi, The oceans' twilight zone must be studied now, before it is too late, Nature 580 (7801) (2020) 26–28, bandiera_abtest: a Cg_type: Comment Subject_term: Ocean sciences, Biogeochemistry, Ecology, Fisheries. doi:10.1038/d41586-020-00915-7.
URL https://www.nature.com/articles/d41586-020-00915-7

[8] L. Eduardo, A. Bertrand, Deep-pelagic ecosystems should be considered as social–ecological systems, Nature Ecology & Evolution 9 (5) (2025) 745–748. doi:10.1038/s41559-025-02692-w.
URL https://www.nature.com/articles/s41559-025-02692-w

[9] "echoview".
URL https://echoview.com

[10] R. J. Korneliussen, E. Ona, I. Eliassen, Y. Heggelund, R. Patel, O. R. Godø, C. Giertsen, D. Patel, E. Nornes, T. Bekkvik, H. P. Knudsen, G. Lien, The large scale survey system - lsss, in: Proceedings of the 29th Scandinavian Symposium on Physical Acoustic, Vol. 29, Ustaoset, 2006, p. 6.
URL https://www.academia.edu/143858735/The_large_scale_survey_system_LSSS

[11] V. M. Trenkel, L. Berger, S. Bourguignon, M. Doray, A. Et., Overview of recent progress in fisheries acoustics made by ifremer with examples from the bay of biscay, Aquatic Living Resources (2009). doi:10.1051/alr/2009027.
URL https://sci-hub.fr/10.1051/alr/2009027

[12] Y. Ladroit, P. C. Escobar-Flores, A. C. G. Schimel, R. L. O'Driscoll, Esp3: An open-source software for the quantitative processing of hydro-acoustic data, SoftwareX 12 (2020) 100581. doi:10.1016/j.softx.2020.100581.
URL https://www.sciencedirect.com/science/article/pii/S2352711020302946

[13] Y. Perrot, P. Brehmer, J. Habasque, G. Roudaut, N. Behagle, A. Sarré, A. Lebourges-Dhaussy, Matecho: An open-source tool for processing fisheries acoustics data, Acoustics Australia 46 (2) (2018) 241–248. doi:10.1007/s40857-018-0135-x.
URL https://doi.org/10.1007/s40857-018-0135-x

[14] W.-J. Lee, V. Staneva, L. Setiawan, E. Mayorga, C. Tuguinay, S. Butala, B. Lucca, D. Lei, Echostack: A flexible and scalable open-source software suite for echosounder data processing, Python in Science Conference (2024) 386–397doi:10.25080/WXRH8633.
URL https://proceedings.scipy.org/articles/WXRH8633

[15] W.-J. Lee, L. Setiawan, C. Tuguinay, E. Mayorga, V. Staneva, Interoperable and scalable echosounder data processing with echopype, ICES Journal of Marine Science 81 (10) (2024) 1941–1951. doi:10.1093/icesjms/fsae133.
URL https://doi.org/10.1093/icesjms/fsae133

[16] K. Wada, Labelme: Image polygonal annotation with python. doi:10.5281/zenodo.5711226.
URL https://github.com/wkentaro/labelme

[17] K. Wada, Labelme: Image polygonal annotation with python. version 6.0.0.
URL https://github.com/wkentaro/labelme/releases?page=3#release-v6.0.0

[18] D. N. Maclennan, P. G. Fernandes, J. Dalen, A consistent approach to definitions and symbols in fisheries acoustics, ICES Journal of Marine Science 59 (2) (2002) 365–369. doi:10.1006/jmsc.2001.1158.
URL https://doi.org/10.1006/jmsc.2001.1158

[19] H. Kunnath, R. Kloser, T. Ryan, IMOS SOOP-BA netCDF Conventions. Version 2.2, Commonwealth Scientific and Industrial Research Organisation, Hobart, TAS, 2018, oCLC: 1042217660.

[20] A. Ariza, A. Lebourges-Dhaussy, D. Nerini, E. Pauthenet, G. Roudaut, R. Assunção, E. Tosetto, A. Bertrand, Acoustic seascape partitioning through functional data analysis, Journal of Biogeography (2022) jbi.145344 citations (Crossref/DOI) [2024-10-15]. doi:10.1111/jbi.14534.
URL https://onlinelibrary.wiley.com/doi/10.1111/jbi.14534

[21] P. Annasawmy, J. Ternon, P. Cotel, Y. Cherel, E. V. Romanov, G. Roudaut, A. Lebourges-Dhaussy, F. Ménard, F. Marsac, Micronekton distributions and assemblages at two shallow seamounts of the south-western indian ocean: Insights from acoustics and mesopelagic trawl data, Progress in Oceanography 178 (2019) 102161. doi:10.1016/j.pocean.2019.102161.
URL https://www.sciencedirect.com/science/article/pii/S0079661118302313

[22] A. Bertrand, Abracos 2 cruise, rv antea (2017). doi:10.17600/17004100.

[23] A. Bertrand, M. Simier, G. V. F. Afonso, G. Alory, A. Ariza, R. Assunção, J.-C. Auguet, F. Baurand, B. Bec, D. Bonnet, L. Cabanez, C. Carré, A. Chaigneau, A. d. L. M. Cocentino, I. A. Cordeiro, A. Costa da Silva, M. Coubard, M. de Melo Junior, J.-C. Molinero, J. Devesa, G. Domalain, L. N. Eduardo, G. Eldin, G. B. Farias, G. Ferreira, G. Figueiredo, T. Frédou, D. X. Garcia, E. Giachini Tosetto, J. Gonzalez, J. Grelet, S. Hillion, A. Justino, M. C. Jales, S. Lacerda, F. Le Loc'h, A. Lebourges-Dhaussy, R. S. Lima, N. Lins-Silva, A. S. Lira, S. M. d. A. Lira, C. Marcolin da Rocha, J. Martins, A. Médieu, C. Melo, P. A. M. d. C. Melo, M. M. Mincarone, M. Monnier, J.-M. Munaron, S. Neumann-Leitão, J. M. Nogueira, M. Oliveira, B. Padovani Ferreira, L. Pelage, D. Point, U. Pinheiro, F. Roubaud, G. Roudaut, P. Rousselot, J. d. L. Santana, R. Schwamborn, M. Silva-Cunha, A. Soares, J. F. Souza Filho, J. Ternon, P. Travassos, G. Vargas, T. Vaske Jr., R. P. S. Campelo, B. Villarins, F. Lucena-Frédou, The abracos dataset: a multidisciplinary marine ecosystem baseline for the western tropical atlantic (2015–2017), Earth System Science Data Discussions (2026) 1–51doi:10.5194/essd-2026-512.
URL https://essd.copernicus.org/preprints/essd-2026-512/

[24] D. A. Demer, L. Berger, M. Bernasconi, E. Bethke, K. Boswell, D. Chu, R. Domokos, A. Dunford, S. Fassler, S. Gauthier, L. T. Hufnagle, J. M. Jech, N. Bouffant, A. Lebourges-Dhaussy, X. Lurton, G. J. Macaulay, Y. Perrot, T. Ryan, S. Parker-Stetter, S. Stienessen, T. Weber, N. Williamson, Calibration of acoustic instruments., Tech. Rep. 326, ICES Cooperative Research Report (2015).
URL https://repository.oceanbestpractices.org/handle/11329/626

[25] A. S. Brierley, Diel vertical migration, Current Biology 24 (22) (2014) R1074–R1076. doi:10.1016/j.cub.2014.08.054.
URL https://www.cell.com/current-biology/abstract/S0960-9822(14)01067-7

[26] S. S. Urmy, K. J. Benoit-Bird, Fear dynamically structures the ocean's pelagic zone, Current Biology 31 (22) (2021) 5086–5092.e3. doi:10.1016/j.cub.2021.09.003.
URL https://www.sciencedirect.com/science/article/pii/S0960982221012434